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Biochar: The nature-based solution to the growing threat of methane emissions

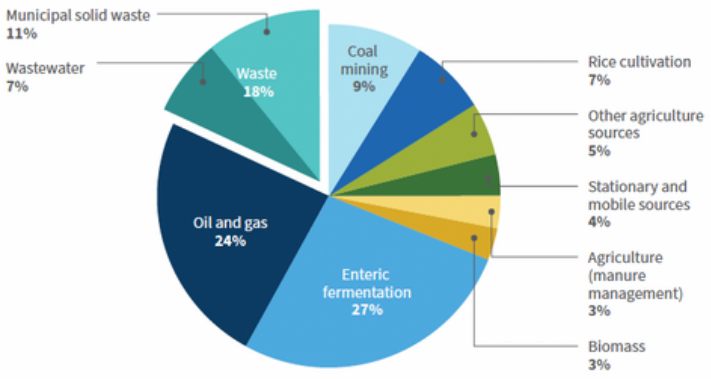
From dairy farms to oilfields to landfills, the charcoal-like substance, best known for its carbon removal capability, is also a proven and highly effective method for keeping methane emissions out of the atmosphere.

During international climate talks in Egypt in November 2022, UN officials joined experts from around the world to issue their most dire warning yet: Time is running out for implementing solutions to stop climate change. Some negotiators, including US Special Climate Envoy John Kerry, focused on reducing methane emissions to meet that goal. “Cutting methane is the fastest opportunity to reduce warming and keep 1.5°C within reach,” Kerry said at COP27. Methane is short-lived compared to carbon dioxide, achieving significant reductions would have a rapid and significant effect on atmospheric warming potential.

Emissions from methane, the world’s second most abundant greenhouse gas after carbon dioxide, must be addressed quickly. The gas, which leaks from oil and gas wells, landfills, melting permafrost, farms, power plants and gas stoves, is more than 25 times as potent as carbon dioxide at trapping heat in its first 20 years in the atmosphere.

As emissions continue to rise, world leaders focus on reducing methane emissions. In 2021, 130 countries, including the US, signed the Global Methane Pledge, which commits them to reducing 2020 levels of global methane emissions by at least 30 percent by 2030. Acting on this pledge, the US, along with the European Union and 11 other countries recently announced plans to speed methane cuts by targeting its largest sources: fossil fuel “super-emitters” such as agriculture, landfills, and oil and gas operations.

Exhibit 1 **Global Anthropogenic Methane Emissions by Source, 2020**



Source: Global Methane Initiative, “Global Methane Emissions and Mitigation Opportunities,” https://www.globalmethane.org/documents/gmi-mitigation-factsheet.pdf

Biochar, a material created by heating biomass (organic fuel such as plants and wood waste) in the absence of oxygen, is an increasingly attractive way to rapidly achieve the greenhouse gas reductions needed to meet the 2030 goal. Over the past few years, biochar has shown tremendous promise as a proven, inexpensive, scalable solution in the race to meet the methane challenge.

Why Biochar Continues to Be a Relevant Climate Technology

The biochar industry is a rapidly growing, global sector, active in multiple markets, including agricultural soil amendment, industrial composting, landscape management, floodplain mitigation, in the built environment, among other sectors. In North America, the biochar market is estimated to be between $50-$100 million, expected to grow by 10X in the next five years. In 2022, the EU had nearly 30 biochar production plants installed, and there are over 50 projects in development in 2023. Within the growing voluntary carbon market, biochar carbon credits made up approximately 80% of delivered credits in 2022.



Photo credit: Syncraft 2023

The practice of making biochar is not new — ancient communities in the Amazon rainforest used biochar as a soil enhancer, and recently researchers, climate advocates and policy makers have explored implementing the technology to help slow climate change. Dozens of studies and pilot projects have shown that biochar can be used to manage methane in a number of important ways, from reducing the amount of the gas released from cow manure to removing methane from landfills.

Biochar is generally produced by heating biomass, such as wood or crop residues, agri-food waste or manures, with little air. While it can be made using low-tech methods, such as charring biomass in pits dug into the ground, producing it at scale involves using advanced systems that capture the heat and gas produced when biochar is created. That heat and gas can then be used to generate electricity or biofuels, or funneled back into the system as a heat source.

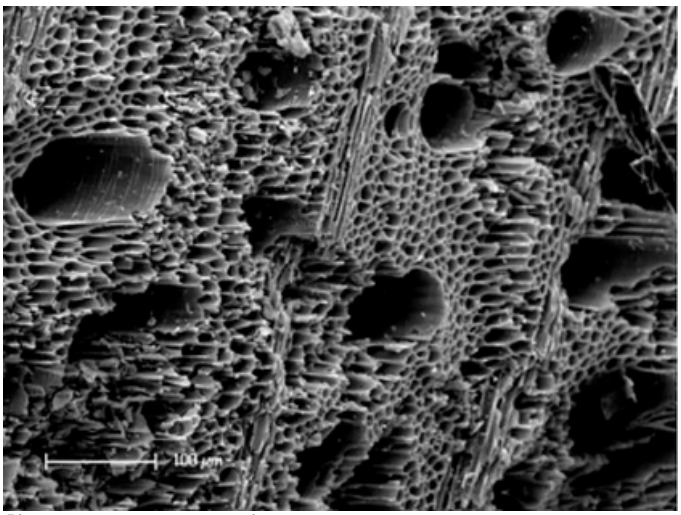


Photo source: www.researchgate.net

Biochar can be highly porous, making it function like a sponge. Its physical and chemical structure allows it to capture and retain moisture, provide a home for beneficial microorganisms, and attach or adsorb nutrients such as nitrogen and phosphorus, as well as gasses including carbon dioxide, methane, and nitrous oxide.

As we learn more about biochar's remarkable

ability to reduce methane emissions, an increasing number of business owners and government entities are turning to the technology to rein in the gas, including the three highest methane emitting sectors.

Biochar on the farm

Biochar can make a significant difference in reducing methane pollution on the farm. Agriculture is a major source of methane emissions, accounting for about 36 percent of the CH4 released in the US, according to US EPA, and about 32 percent worldwide, the United Nations Environment Program estimates.

Ruminants such as cows, sheep and goats are some of the biggest culprits — primarily due to fermentation during the digestive process and to manure management practices. Adding biochar to manure composting has proven to be a highly effective way to cut methane pollution from dairy operations. A 2022 study found that composting manure with biochar reduced methane by 79 percent compared to composting without it.

But methane in the agriculture sector doesn't just come from livestock. Rice cultivation, one of the most commonly grown crops in the world, involves flooding fields, creating perfect conditions for methane-emitting bacteria. Rice cultivation represents about 12 percent of global methane emissions, and 1.5 percent of all global greenhouse gas emissions. Biochar has shown some promise for harnessing those emissions too, particularly biochar made from rice straw. A 2011 analysis that added straw char to the soil in rice paddies found a 91 percent reduction in methane releases. How well biochar tamps down methane emissions from rice cultivation over the long term under different soil and weather conditions, though, remains a topic for future research to explore.

Limiting methane from landfills

Hidden in discarded banana peels and clumps of wilted lettuce are potential methane emissions just waiting to be released. Each year more than 35.2 million tons of food waste ends up in landfills, where it releases methane as it decomposes. Emissions from municipal landfills — the third largest source of methane in the US — amounted to about 109.3 million metric tons of carbon dioxide equivalent in 2020. Globally, food waste accounts for about 8 percent of greenhouse gas emissions.

Wood waste makes up additional solid waste that is disposed of at the landfill. A more efficient way to reduce methane from landfills — particularly in areas where there is timber production, as well as wildfire mitigation — is to divert the wood waste and turn it into biochar.

Landfill operators often bury the dumping ground with a thick layer of soil. Researchers have found that mixing biochar into this soil cover reduces methane emissions more than soil alone. That's because microbes that use methane as food thrive in biochar's nooks and crannies, a 2022 analysis found. Specially designed biochar that is already stocked with methane-consuming microbes called methanotrophs, as opposed to biochar that is colonized by them over time, is even more effective in trapping the gas, according to a 2019 study.

Methane is pervasive in "capped" oil and gas wells

Oil and gas production continues to be the largest source of methane emissions in the US, accounting for one-third of total releases. Currently, the Biden administration is tightening regulations in the sector as it seeks to make good on its promise to curb methane emissions. President Biden recently announced a pair of new rules that together aim to cut methane pollution from existing oil and gas wells by more than 85 percent.

Part of the industry's methane problem lies in the millions of old, tapped-out wells that languish across oil and gas fields around the world. Long after a well is depleted, methane still escapes. There are about 29 million of these old oil and gas wells globally, and more than four million in the US. The wells release about 2.5 million tons of the gas each year.

A series of pilot projects in the US have indicated that biochar-enhanced concrete is an effective way to cap abandoned oil and gas wells, which can emit significant amounts of methane. Remediation of these wells typically involves plugging them with concrete, which can crack over time, and is a source of emissions itself. Pilot projects are still in the early stages exploring biochar as an additive to the concrete, helping to ensure the methane stays underground. Because of its porous structure and persistence, biochar may also make concrete well plugs better able to withstand earthquakes. Several recent studies found that adding even a small amount of biochar to cement is enough to increase the strength by 20 percent to 78 percent, depending on the type of feedstock used to make the biochar.

In all three of these key sectors — agriculture, solid waste management and oil and gas production — biochar can make a significant difference in reducing methane emissions while providing carbon sequestration opportunities. Research and industry experience show that biochar is a cost-effective, proven, and scalable solution to reducing greenhouse gas emissions, including methane.

Join the International Biochar Initiative in putting biochar to work for the climate. Contact IBI to learn more about how biochar can support your methane emissions reduction work.

About the International Biochar Initiative:

The International Biochar Initiative (IBI), the world's global-facing biochar association, is a membership-based organization that serves as a platform for the exchange of information and activities in support of biochar research, development, demonstration, and commercialization. IBI has over 600 members based in 63 countries, including researchers, commercial entities, policy makers, development agents, farmers, and others committed to supporting sustainable biochar production and utilization systems that remove carbon from the atmosphere using circular economy solutions.

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Additional Resources

For further information about this topic, organized by paper sections

Introduction:

"Global Methane Assessment: Benefits and Costs of Mitigating Emissions" UNEP and Climate & Clean Air Coalition

"Importance of Methane" EPA

2018 IPCC Special Report on 1.5 degrees C of global warming

2019 IPCC Special Report on Climate Change and Land

IPCC 5th Assessment

Delivering on the US Methane Emissions Plan, Nov. 2022

Overview of biochar:

"Assessing the diverse environmental effects of biochar systems: An evaluation framework" *Journal of Environmental Management*

"Biochar-based carbon sinks to mitigate climate change" EBI Whitepaper

"Soil carbon sequestration and biochar as negative emission technologies" Global Change Biology

Agriculture:

"Overview of Greenhouse Gases — Methane Emissions" EPA

"Agricultural methane emissions and the potential for mitigation" Philosophical Transactions of the Royal Society A

"How can biochar in livestock feed reduce methane emissions?" Simon Burgess, Reaseheath College TED talk

"A systematic review of biochar use in animal waste composting" *Waste Management*

"Biochar for modification of manure properties" *Advances in Chemical Pollution, Environmental Management and Protection*

"Dairy Manure Co-composting with Wood Biochar Plays a Critical Role in Meeting Global Methane Goals" *Environmental Science & Technology*

Rice paddies:

"Effects of biochar on methane emission, grain yield, and soil in rice cultivation in Thailand" Carbon Management

"Exploring long-term effects of biochar on mitigating methane emissions from paddy soil: a review" Biochar

Landfills:

"Comparison of the methane-oxidizing capacity of landfill cover soil amended with biochar produced using different pyrolysis temperatures" *Science of the Total Environment*

"Effects of biochar amendment on geotechnical properties of landfill cover soil" *Waste Management & Research*

"Effects of biochar on bacterial communities in a newly established landfill cover topsoil" *Journal of Environmental Management*

"Enhanced Landfill Methane Oxidation Using Activated Biochar" ASCE Geo Congress 2022

"Estimates of solid waste disposal rates and reduction targets for landfill gas emissions" *Nature Climate Change*

"Methane Emission Reduction Enhanced by Hydrophobic Biochar-Modified Soil Cover" *Processes*

"Methane Emission Reduction and Biological Characteristics of Landfill Cover Soil Amended With Hydrophobic Biochar"

Oil and gas:

"Biochar amendment as a remediation strategy for surface soils impacted by crude oil" *Environmental Pollution*

"Methane emissions from abandoned oil and gas wells in California" *Environmental Science & Technology*

"Biochar use in Capping Abandoned Oil and Gas Wells" International Biochar Initiative